We present a robotic model of "attachment"—the primary bond between child and caregiver that shapes relationship behaviour throughout our lives. Following a dynamical systems approach, we model attachment as a behavioural coupling between motivational oscillators and report results using both computer simulation and a robotic dyad. We find that this oscillator framework generates rich patterns of behaviour, in both the simulated and embodied settings, that can be associated with observations of parent-child interactions in the "strange situation" procedure, an experimental paradigm that is widely studied in human relationship science. We also find that parameters within the model that modulate coupling can be set to capture some of the characteristic features of human secure, avoidant and ambivalent attachment styles. Finally, using the robotic model, we show that measures of physical and emotional distance (a psychological variable), inferred from sensory data, can serve as effective control parameters for attachment behaviour.

Modelling Human Attachment using Coupled Oscillators in a Simulated and Embodied Robot Dyad

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1 Introduction

The attachment bond—formed between an individual who requires care and another who provides it [1]—is a foundational form of human bonding that has emerged from the mammalian trait of caring for offspring [2]. In this contribution we present a dynamical systems account of the attachment bond, viewed as a form of coupled oscillation, and explore its potential for understanding caregiver-child relationships through computer simulation and in an embodied robotic model. We find that this framework generates rich patterns of robot behaviours that can be associated with quantitative and qualitative observations of the "strange situation" procedure [3, 4], an experimental paradigm that is widely studied in human relationship science.

In the first part of this contribution we discuss human attachment from a motivational perspective and introduce our dynamical systems approach relating it to

previous modelling efforts. We then present a detailed account of our coupled oscillator model followed by theoretical and simulation results showing its ability to capture some of the characteristics features of human attachment behaviour. Our analysis shows how secure, avoidant, and ambivalent styles, as studied in the human attachment literature can emerge as phenomena of oscillator synchronization and anti-synchronization. We next explain the embedding of the coupled oscillator model within the control systems of a dyadic robot pair before describing experiments, first in simulation and then using two small robot platforms (the Miro-e robot [5]), that model simplified child/-caregiver interactions. The discussion summarises our key findings and considers their relevance to the understanding of human attachment. Finally, we briefly consider how the ability to match human attachment style and to generate appropriately-matched robot behaviours could aid the design of effective and appropriate social behaviour for robots. The current article builds on and significantly extends an earlier pilot study [6].

2 Attachment as a motivational system

Attachment theory, developed by John Bowlby in the 1950s and 1960s [1] and expanded by Mary Ainsworth [3], highlights the importance of primary relationships, particularly that between a child and its caregiver (typically the mother), as being critical in establishing ways of relating to others that persist through life. The classical theoretical approach uses different categories to classify attachment styles in children, as revealed by a laboratory procedure termed the "strange situation" procedure (SSP) [3, 4]. In a typical SSP experiment, a child and their caregiver are observed during multiple short episodes during which the caregiver is either absent, present alone, or present with a stranger. A child that is observed to have a weak emotional bond with the caregiver is characterised as having an *avoidant* attachment style and one that appears preoccupied about the caregiver's availability is characterized as *ambivalent*. A key dependent variable is the physical distance between child and caregiver, as illustrated in figure 1—a *securely* attached child typically shows bouts of exploratory behaviour, that involve moving away from, then returning to, the caregiver, on a quasi-periodic basis. In contrast the ambivalent child will typically stay close to the caregiver while the avoidant child will explore while paying little attention to the caregiver.

More recent work has proposed that the attachment styles identified by Bowlby and Ainsworth are best considered as emerging from a multi-dimensional motivational system [8, 9]. For instance, Gagliardi [9], describes a 7-dimensional framework with three basic dimensions—avoidance, ambivalence and disorganisation—viewed as emerging during an early "imprinting" period (6 to 24 months). According to this approach, avoidance has an essentially emotional nature, meaning that it is primarily concerned with the affective bond between child and care-giver. Ambivalence, on the other hand, has a more situational or physical nature, meaning that it is concerned with the caregiver's availability. Building on this analysis, Gagliardi [10] developed an information-theoretic model in which the child and caregiver were modelled as point agents in a two-dimensional space inspired by the SSP. The child's behaviour, derived from its attachment motivational system, was modelled as situated in the

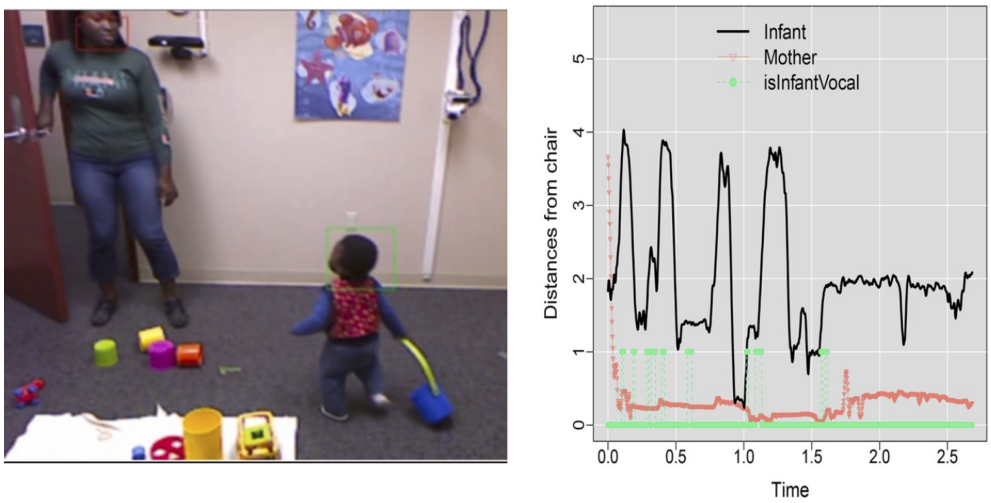


Fig. 1 Left. an SSP room with a mother (caregiver) and her child. Reprinted from [4] with permission from Elsevier. **Right. Example behavior patterns displayed in the SSP**. Plots show distance of the child (black line) and mother (red line) from the chair during a reunion episode and illustrate the oscillatory movement patterns often seen with securely attached children. Reproduced from [7] with permission from Springer Nature.

two-dimension avoidance-ambivalence space. The caregiver, whose behaviour was the manifestation of a care-giving motivational system, was modelled as situated in a two-dimensional sensitivity-responsiveness space. Simulation experiments with this model generated alternating patterns of exploration and approach behaviour that were aligned with the attachment literature confirming the utility of the dimensional framework.

2.1 A dynamical systems approach

In originating attachment theory, Bowlby [1] was concerned with applying insights from both ethology and cybernetics (control theory)—key scientific movements at the time he was working—to the understanding of human personality and relationships. A number of subsequent models have built on Bowlby's enthusiasm for computational accounts (see [11]), most notably, Petters [11], who developed a cognitive architecture for agents to implement and test aspects of attachment theory, and Gagliardi [10], who more recently developed the information-theoretic account described above motivated by the emerging dimensional view. The current study complements this work by focusing on the dynamics of attachment behaviour, through a relatively minimal coupled oscillator model, an emphasis that might have appealed to Bowlby's search for concise explanations that can capture regularities underlying observable human behaviour as we set out below. For instance, in his major treatise on attachment, and with specific regard to a control theoretic perspective, Bowlby wrote:

"Anyone who observes how a mother and her one- or two-year-old child behave over a period of time will see that each exhibits very many different patterns. Whilst some

of the behaviour of each partner has the effect of increasing or maintaining proximity between the pair, much of it is of a completely other sort. [...] Thus there is a dynamic equilibrium between the mother–child pair. Despite much irrelevant behaviour by each, and some competing and some incompatible or contrary behaviour, distance between them is as a rule maintained within certain stable limits. To understand how this comes about it is useful to regard the spatial relationship between the two as the resultant of behaviour of the following four classes: (a) the child's attachment behaviour; (b) behaviour of the child that is antithetic to attachment, notably exploratory behaviour and play; (c) the mother's caregiving behaviour; (d) behaviour of the mother that is antithetic to parental care. [...] Each class of behaviour, moreover, is likely to be affected by the presence or absence of the others, because the consequences of behaviour of any one class are likely either to elicit or to inhibit behaviour of the other three classes. Thus, when mother is called away a child's attachment behaviour is likely to be elicited and his exploratory behaviour inhibited; conversely, when a child explores too far, a mother's care behaviour is likely to be elicited and whatever else she is doing to be inhibited. In a happy couple these four classes of behaviour occur and progress together in harmony." ([1], ch. 13)

In the current work we follow a dynamical system perspective [12–15] modelling the child-caregiver relationship, very much as described above by Bowlby, as being composed of two coupled systems—a child that is motivated to attach and a caregiver that is motivated to provide care—both seeking to balance other motivations such as to explore. This model also builds on the dimensional view of avoidance and ambivalence, and the emphasis within the Gagliardi model [10] on physical and emotional distance. However, our approach simplifies the information-theoretic model of Gagliardi into a set of differential equations that are more amenable to dynamical analysis while still capturing key characteristics of the dimensional view. We also extend the modelling approach beyond investigating point agents in two-dimensional worlds, by developing a robotic implementation that operationalizes additional elements of the Gagliardi model that assume the capacity of the two agents to infer each other's attachment state.

3 The Model System

Our dynamical model stems from the motivated-behaviour-as-an-oscillation framework [15, 16] in which the internal needs of the agents shape a dynamical landscape in a manner that can generate or extinguish attractors in the phase space. From this point of view, different motivational systems are activated depending upon the location of the agent's internal state in such a phase space by using attractor-centered *readout* functions as further explained below. The results presented here are a natural corollary of such approach—if motivation is an oscillation, interaction becomes coupling. The coupling of dynamical oscillators has been widely studied (see [17]). For instance, Helm et al. [19] used a couple-oscillator model to understand interpersonal synchrony with respect to physiological signals such as heartbeat and respiration (see [20]). The current work extends this methodology to the domain of interpersonal behavior using attractive and repulsive coupling to simulate ambivalent and avoidant tendencies. This

kind of coupling generates a wide range of behaviours including inhomogenous steady states (IHSS), oscillation death, synchrony, anti-synchrony and bi-stability [18].

In [16] we modelled motivational dynamics as a particle, here denoted x, moving within a one-dimensional motivational space so as to minimise energy (see also [15]). For two motivations, we define the potential $\Psi(x)$ such that:

$$\Psi(x,u,v) = \frac{1}{2}\left((x+1)^2(x-1)^2 + (x+1)^2\frac{(1-u)}{2} + (x-1)^2\frac{(1-v)}{2}\right), \quad (1)$$

Here u and v are two needs corresponding to the two motivations, these could relate to physiological variables, such as hydration or blood sugar level, or to more psychological constructs such as emotional needs. Equation (1) models an interaction between a fast gradient dynamics for the motivational state, x, and a slower accumulation of the needs (see [14] for an introduction to fast-slow analysis of multi-timescale systems). This system will evolve over time to minimise energy by following the negative gradient of the potential (see [15] for further explanation and background).

To model attachment, we assume each agent has two motivations: to receive or give care, and to explore. To simplify the model, we assume that both motivations are encoded by the same underlying need $u = -y$ and $v = y$. This remodels attachment as an approach-avoid conflict. Replacing in equation (1) we get:

$$\Psi(x,y) = \frac{x^4}{2} - \frac{x^2}{2} + xy + \frac{1}{2}.$$

Note that when $y = 0$, this potential has two minima (point attractors), however, as y becomes positive or negative, one of the minima disappears allowing the system to settle on the other one as illustrated in Figure 2).

The slow need accumulation is modelled as

$$\dot{y} = bx + \text{coupling terms}.$$

We showed previously [16] that such a system generates relaxation oscillations therefore we obtain two coupled Van der Pol oscillators represented by the system:

$$\begin{aligned}
\dot{x}_c &= -\frac{\partial \Psi}{\partial x_c}(x_c, y_c) = x_c - 2x_c^3 - y_c, \\
\dot{y}_c &= bx_c - \varepsilon_{Am} f(y_p, y_c) - k_0(\varepsilon_{Av} g(y_p, y_c) + \delta), \\
\dot{x}_p &= -\frac{\partial \Psi}{\partial x_p}(x_p, y_p) = x_p - 2x_p^3 - y_p, \\
\dot{y}_p &= bx_p + \varepsilon_{Am} f(y_p, y_c) - k_0(\varepsilon_{Av} g(y_p, y_c) + \delta).
\end{aligned} \quad (2)$$

Here x_i, where $i \in \{c, p\}$ (c representing child and p representing parent/caregiver), is the current motivational state based on receiving or giving care, and the variable y_i, where $i \in \{c, p\}$, is the accumulated need. ε_{Am} and ε_{Av}, referred to below as the *ambivalence* and *avoidance* terms respectively, are parameters that weight the

contributions of the two coupling functions and that link the accumulated needs of the two agents. These are defined as:

$$\begin{aligned} f(y_p, y_c) &= y_c + y_p \\ g(y_p, y_c) &= y_c - y_p. \end{aligned} \tag{3}$$

The functions f and g are inspired by the "repulsive" and "attractive" coupling used in [18]. Note that ε_{Am} and ε_{Av} are parameters of the caregiver-child dyad. k_0 and δ are chosen so that the parameters change in the interval $[0, 1]$ in an interpretable way. The dynamics of these caregiver-child dyad parameters and how the model behaves will be explained further below.

4 Dynamics of the Disembodied Model

Figure 2 shows the phase space geometry under different coupling regimes for the caregiver and the child. In all the figures, we show the nullclines for each oscillator (a nullcline is a curve in the phase plane where one of the variables undergoes no change). In the absence of any coupling, the two nullclines intersect in a unstable fixed point and a relaxation oscillation emerges as an attractor for the system (figure 2, left).

In the fully ambivalent scenario, the ambivalence term will push the two oscillators apart by the repulsion mechanism (figure 2, center). On the other hand, the avoidance term will draw them together towards a common fixed point which corresponds to a state of perpetual exploration (figure 2, right).

We now proceed to analyze the full system as the ambivalence and avoidance parameters change.

4.1 Ambivalent regime

We first explore the ambivalent regime by varying ε_{Am} and fixing ε_{Av} to be zero. This system has 2 fixed points, the trivial one at the point $(0, 0, 0, 0)$, which is unstable for Van-der Pol oscillators, and an in-homogenous steady state (IHSS) at $FP = (\bar{x}, \bar{y}, -\bar{x}, -\bar{y})$, with

$$\bar{x} = \pm\sqrt{\frac{\varepsilon_{Am} - b}{2\varepsilon_{Am}}} \tag{4}$$

$$\bar{y} = \pm\frac{b}{\varepsilon_{Am}}\sqrt{\frac{\varepsilon_{Am} - b}{2\varepsilon_{Am}}}, \tag{5}$$

which is obtained by setting $\dot{x}_i = 0$ and $\dot{y}_i = 0$, and solving the resulting quadratic equation. The later is of interest to us. It can be observed that, as ε_{Am} increases, there is a saddle-node bifurcation. In the ambivalence-only case, the resulting system can be proved to be bounded for $0 \leq b \leq 1$ using as a Lyapunov function the euclidean distance of the the oscillator states from the origin. In this case, the disappearance of the fixed point at $\epsilon_{Am} < b$, guarantees the existence of an stable limit cycle.

At $\varepsilon_{Am} = b$ the fixed point (s) comes into existence but it remains unstable for a small range of values displaying chaotic behaviour, as we will show later, before

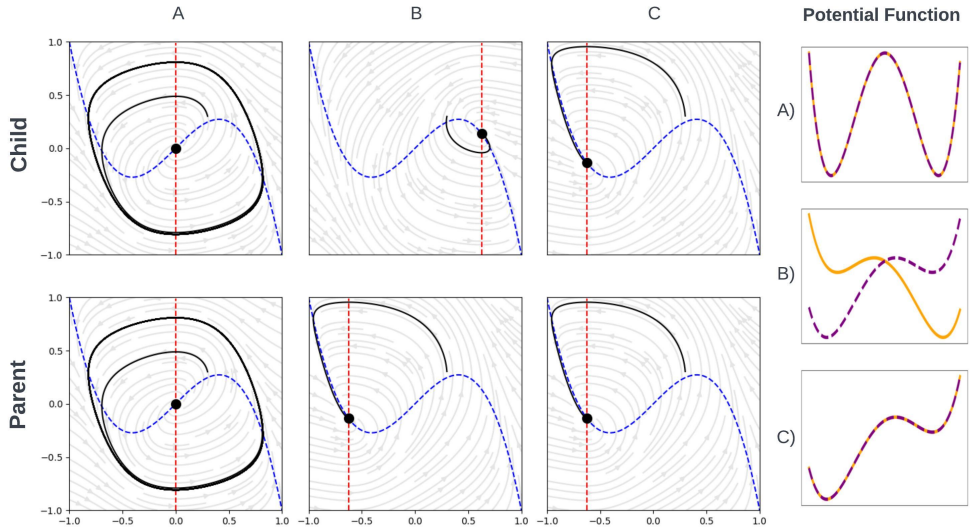


Fig. 2 Illustration of the mechanisms of attachment in the coupled oscillator model. We assume $f(x, y) = g(x, y) \equiv k$ to illustrate the attraction and repulsion effects for illustration purposes. We show the solution to (2) as a closed trajectory in the xy phase space for the child (top row) and for the caregiver (bottom row). The blue curve corresponds to the nullcline $\dot{x} = 0$, th red line is $\dot{y} = 0$. The black dot is a fixed point of the system. **A**. The first column shows the behaviour of the uncoupled oscillators and illustrates the existence of closed trajectories (i.e. relaxation oscillations). **B**. Shows the putative ambivalent behaviour with $\epsilon_{Am} = 1$ and $\epsilon_{Av} = 0$. Notice that the system is attracted to opposite fixed points that correspond to perpetual exploration and approach. **C**. Putative avoidant behaviour, with $\epsilon_{Av} = 1$ and $\epsilon_{Am} = 0$, showing how both agents are attracted to an exploration fixed point. Right column shows the corresponding potential functions $\Psi(x)$ for the child (purple) and for the parent/caregiver (yellow) at the fixed points.

settling in an ambivalent steady state in which the needs are opposite to each other. We choose $b = 0.5$ to ensure that the system reaches the steady state in the range $\epsilon_{Am} \in [0, 1]$. Figure (3) shows the dynamics of the model as the ambivalence parameter increases in the interval $[0, 1]$, (3b) shows the route to chaos just described illustrated by the Imaginary part of the fixed point.

This route to chaos can also from the perspective of relaxation oscillations described in figure 2. As the ambivalence changes, the nullcline $\dot{y_c}, \dot{y_p} = 0$ approaches the fold points of both oscillators. As it approaches the fold, the oscillators spend more time in one of the branches. This is characterised by a slow-down in the frequency of oscillation (figure 3). When it crosses the fold, the new fixed point appears because now the stable branch of the critical manifold is intersecting this nullcline. The subsequent emergence of chaos in the oscillation quenching is known as Oscillation Death (OD).

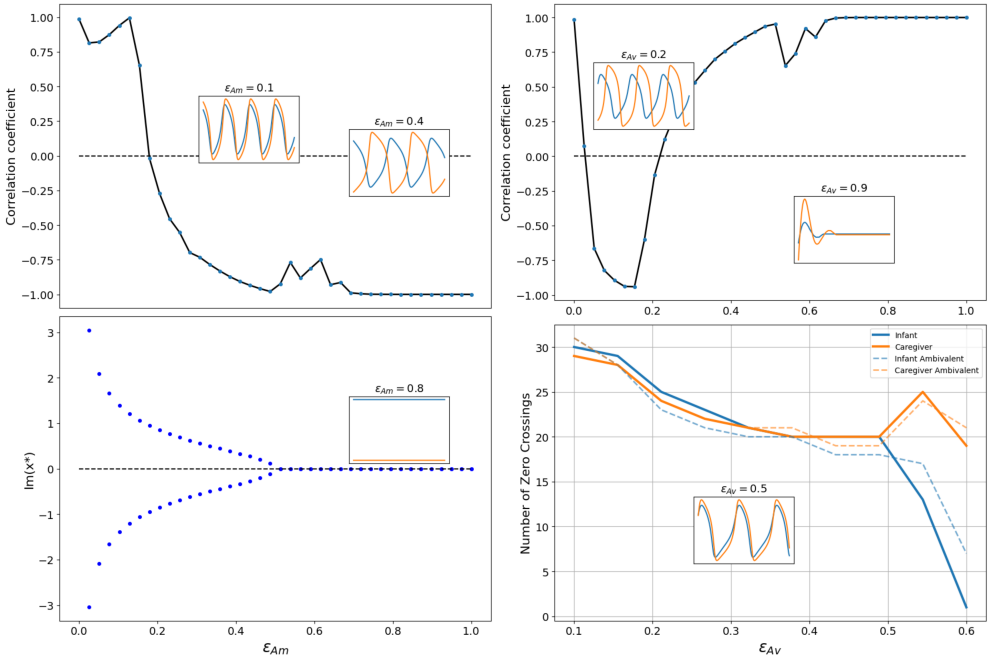


Fig. 3 **Top left**. Correlation coefficient of the needs of the parent and the child as the **ambivalence** parameter ϵ_{Am} increases. For lower values, the dyad is synchronized (inset at $\epsilon_{Am} = 0.2$). As the parameter increases, it becomes steadily desynchronised until it collapses into an inhomogenous steady state in which the parent is not able to satisfy the needs of the child. Note that as ambivalence increases there is a slowing down of the rate of oscillation (compare insets at $\epsilon_{Am} = 0.2$ and 0.4). **Bottom right. Route to chaos.** A limit cycle loses stability and a steady state appears as the parameter increases. Shown in the graph is the imaginary part of the roots in equation 5. **Top right**. The correlation coefficient in the **avoidant** regime drops quickly for small values of the parameter. This region is characterized by the needs of the child tracking the needs of the parent. Half way through the interval, the system becomes synchronized just before settling into the only steady state: one of continuous exploration or complete avoidance. **Bottom right**. Frequency estimated by the number of zero crossings as the avoidant and ambivalent parameters are increased towards the bifurcation point.

In order to understand the interaction between the needs of the caregiver-child dyad, we compute correlation coefficient between the two needs:

$$r(x_c, x_p) = \frac{\sum_i (x_c^i - \bar{x_c})(x_p^i - \bar{x_p})}{\sqrt{\sum_i (x_c^i - \bar{x_c})^2}\sqrt{\sum_i (x_p^i - \bar{x_p})^2}}. \tag{6}$$

The dyad starts synchronized but promptly becomes desynchronized and then anti-synchronized as the parameter increases in the oscillatory regime; when the child needs care, the parent is not available to provide it.

4.2 Avoidant regime

We next fix ε_{Am} to zero and explore the avoidance regime. This system has a unique fixed point at $(\bar{x}, \bar{x}, \bar{y}, \bar{y})$; the dyad tend towards the same steady state. This state is given by the solutions of the cubic equation:

$$x^3 + \frac{1-q}{2q}x + \frac{\delta}{2} = 0,$$

with $q = \frac{k_0 \epsilon_{Av}}{b}$. We have chosen k_0 and δ such that

$$-\Delta = -4\left(\frac{1-q}{2q}\right)^3 - 27\left(\frac{\delta}{2}\right)^2 < 0,$$

for all $\epsilon_{Av} \in [0, 1]$, so that there is only one fixed point. Δ is the discriminant obtained from solving the depressed cubic equation. Similar to the ambivalent case, this fixed point becomes stable as the parameter increases, however, in this case, it is shared by both agents in the system. Figure (3c) shows the dynamics of the correlations in the avoidant dyad.

Further details about the dynamics of the dyad can be obtained by computing the relative phase between the needs of the child and the caregiver as illustrated in figure 4. The blue regions correspond to periodic behaviour that includes synchronization, anti-synchronization and phase locking. A synchronization region (pink band in the inset) corresponds to putative secure states. For the ambivalence(avoidance)-only regines studied in this paper, the relative phase tends to increase for both avoidance and ambivalence until it reaches a chaotic region that is part a larger region of the parameter space of non-periodic behaviours.

This chaotic behaviour is shown in detail in figure 4 (right) by means of a bifurcation diagram: a small parameter range separates stable need satisfaction from extreme avoidance and ambivalent attachment behaviours. This region is composed of different bands of non-periodic (chaotic) behaviour, along with periodic windows; we associate this with disorganise attachment.

5 Robotic Implementation

Embodied (robotic) modelling allows theoretical proposals to be challenged in a real-world setting that forces more of the assumptions of the approach to be cashed out in mechanistic terms [15, 21]. The added complexity of the physical world also provides additional tests of the sufficiency and completeness of the embedded model [22]. To test our coupled oscillator framework we use the MiRo-e robotic platform [5]. In previous studies, we have demonstrated control of this platform with a brain-inspired layered cognitive architecture [23], and have explored the robot's capabilities in affective communication [24]. Here we implement the attachment dynamical model in a robot-robot dyad, where one robot assumes the role of the parent/caregiver and the other the role of the child in a simplified model of the SSP as explained next.

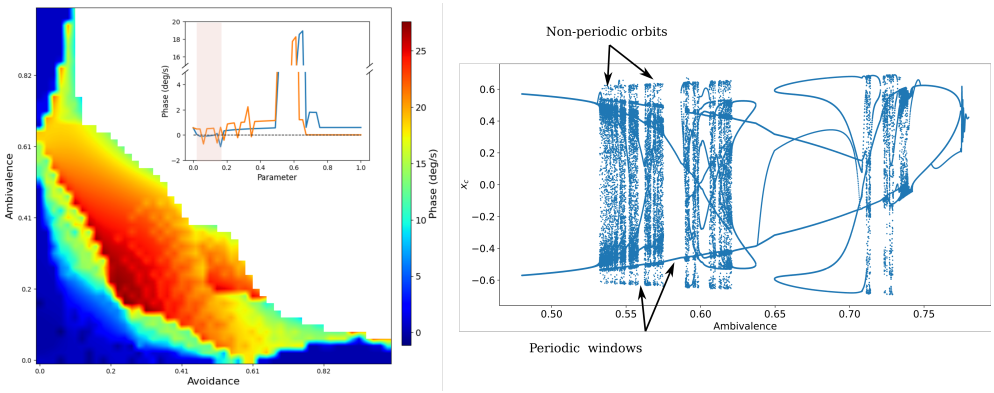


Fig. 4 **Left. Average relative phase** between the need states of the child and caregiver for different values of the avoidance and ambivalence parameters, from a fully anti-sychronised initial condition. The white regions are combinations of parameters for which the model blows up. The red and green regions correspond to chaotic and non-periodic phenomena. Blue regions correspond to oscillations with two states (approach and avoid). In this work we focus on the strips $\epsilon_{Av} = 0$ and $\epsilon_{Am} = 0$ shown in the inset, here the blue/orange lines show how relative phase changes as the $\epsilon_{Am}/\epsilon_{Av}$ parameter is increased with other held at zero. Relative phase is measured in degrees per second. **Right. Period doubling and chaotic regimes** illustrated by a detailed bifurcation diagram computed by capturing the crossings of a Poincare section defined at $y_c = 0$.

5.1 Action selection

To select actions, we use readout functions that indicate whether the approach or explore behavioural systems should be activated. When the motivational state is positive, the approach behavioural system controls the robot, and when it is negative, the exploration system takes over. The architecture of the system along with the dynamics of the caregiving interactions are shown in figure 5 and explained next.

Mathematically, we express the activation level of each system as:

$$A_{\text{approach}}(x_i) = \Theta(x_i), \tag{7}$$

$$A_{\text{explore}}(x_i) = \Theta(-x_i), \tag{8}$$

where $\Theta(\cdot)$ is the Heaviside function (a form of step discontinuity). Following [10], we define a way of influencing the other agent's motivational state by the perceived physical (d_p) and emotional distances (d_e) between them. To achieve this, we modify equations in (9) to include those behavioural estimates:

$$\begin{aligned} f(y_p, y_c) &= y_c + y_p \pm d_p \\ g(y_p, y_c) &= y_c - y_p - d_e. \end{aligned} \tag{9}$$

Therefore, the physical distance influences the ambivalent coupling while the emotional distance influences the avoidant coupling. The emotional distance (d_e) increases over

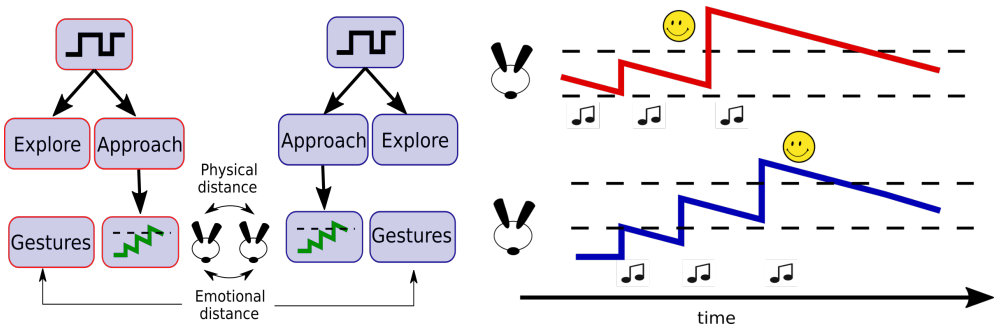


Fig. 5 Left. Cognitive architecture of the child and caregiver robots. The oscillator controller acts as a motivational switch that activates one of the two parallel behavioural systems—approach and explore—depending on need state. **Right. Evidence accumulation for the caregiving interaction**. The process has two thresholds. The first one initiates vocalizations for each agent. Once the accumulation is above the first threshold, the agent will respond with a new vocalization each time it detects a potential response. The process finishes after crossing the second threshold at which point the emotional distance is reset to 0.

time following the first order equation:

$$\dot{d}_e = \kappa_e(1 - d_e), \tag{10}$$

when there is no care being given, and is reset to $d_e = 0$ once a care-giving interactions is successful. κ_e is the rate of accumulation of the emotional distance. The emotional distance here is considered as emotional connectivity, which we treat here as similar to physiological needs such as thirst or hunger. This need is considered to be satisfied through interactions such as care-giving interactions.

The physical distance is the normalized euclidean distances from the parent:

$$d_p = \min\left(1, \frac{d_x(p_c, p_p)}{L}\right), \tag{11}$$

where $d_x = \|p_c - p_p\|_2$, p_c and p_p are the position vectors of the child and the parent, estimated from odometry data, and L is the size of the arena.

In this work, both agents have access to the internal accumulated needs of the other agent as required for the coupling of the oscillators, therefore the effects of d_p and d_e are only modulatory. Future work will focus on the estimation of these variables from behaviour. Here, we consider responsiveness and availability to the other agent as a measure of physical distance, thus we base this calculation on the caregiver and child's respective positions.

5.2 Giving care

The caregiving interaction happens by means of vocalizations performed using MiRo's biomimetic voice synthesis [25]. Each agent accumulates evidence about the interaction

using a simple drift diffusion process illustrated in figure 5 (right):

$$dX_i = (-aX_i + I) + \sqrt{\sigma}dW, \tag{12}$$

where a is the rate of accumulation and σ the variance of the random process. The parameters of the stochastic equation are chosen empirically to match the rates and time scale of the oscillation. The input I is defined through a simple template matching algorithm: each robot's audio input is continuously sampled by the approach motivational system for energy peaks in the preferred frequency bands of each agent. Those bands correspond to the bands and harmonics where the biomimetic vocalization is more likely to occur depending upon the mass of the animal being simulated [25].

Because both the vocalization and the environment are noisy, the caregiving interaction is only complete after enough evidence is accumulated following the procedure described in figure 5 (right).

The caregiving interaction for the caregiver has been designed to start from the first threshold where it simultaneously makes and expects audio responses until it crosses a threshold where it deems the child's responses satisfactory. In the case of the child, it will begin from below the first threshold, and either randomly drifts towards it or crosses over it when it receives enough stimulation from the caregiver.

6 Towards a Robot Model of the Strange Situation Procedure

6.1 A biased random walk model

We first evaluate the dynamical effects of the model on the behaviour of a simulated robot dyad (child, caregiver) in a 2D environment and with simplified approach/explore behaviour. At each iteration of the simulation, each agent chooses a direction towards the other if the state corresponds to *approach*, or a random position in the environment if the model's state is *explore*. The movement therefore evolves as a biased random walk. To study the emergent behaviours associated with these regimes we fixed exploration sites at locations $(1, 1)$ and $(-1, -1)$ for the caregiver and the child respectively. In the following experiments we vary either the avoidance (ϵ_{Av}) or the ambivalence (ϵ_{Am}) parameter across the range 0.1-0.9 while holding the other parameter constant at zero as in the disembodied model. To simplify the discussion we assume that any regime displaying synchrony is a secure state.

6.1.1 Secure-to-avoidant transition

As previously noted (figure 3) the dynamical dyad experiences different regimes as the avoidance parameter is increased. Figure 6 shows that, in the simulated robotic model, even if with small non-zero values of the avoidance parameter (leftmost panels) there is some de-synchronization and this increases with larger values up to 0.5. In terms of behaviour, de-synchronisation has the effect that the agents visit each other's exploration bases intermittently. As avoidance increases above 0.5 (right panels) there is a short period of synchrony and the two oscillators then become phase locked before

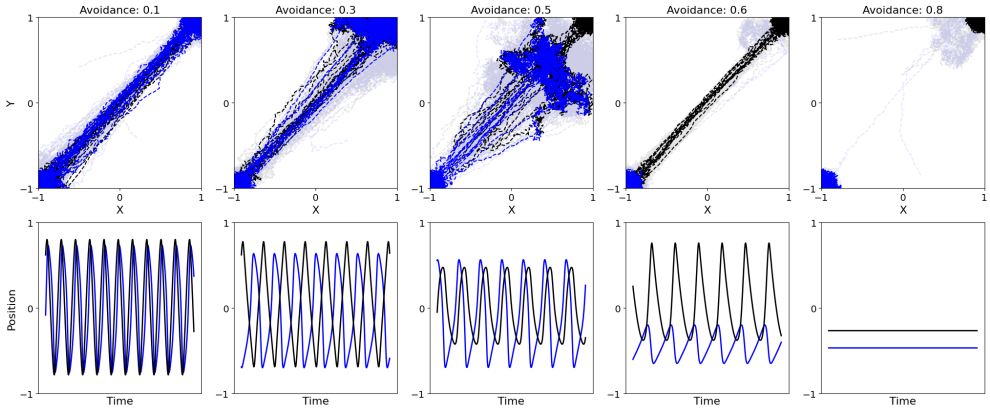


Fig. 6 Secure-to-avoidance transition. Top panels show the trajectory of two simulated agents in a 2D environment (blue: child, black: caregiver). Each has an exploration base at the corners of the square arena. The simulation is run for 100000 steps but only the last 10000 positions are shown (early trajectories are faded). Bottom panels show the needs of the child and the caregiver as a function of time. The value of the avoidance parameter ϵ_{Av} increases from left to right. $\epsilon_{Am} = 0$.

transitioning into a short chaotic regime in which the amplitude and frequency of the transitions become unpredictable. After this, the states start becoming more negative until both settle in a perpetual exploration state and the oscillation is quenched.

6.1.2 Secure-to-ambivalent transition

Figure 7 shows the behaviour of the simulated agents as the ambivalence parameter ϵ_{Am} increases. In the weak coupling regime ($\epsilon_{Am} = 0.1$, left panel), both agents are phase locked in a way that generates a fixed meeting point in space (left panel, top). As ambivalence increases, the system transitions to a de-synchronized state before a more pronounced period of chaotic or "putative disorganised" attachment (ambivalence= 0.6). In figure 8 we take a closer look at this transition, note how the unpredictable nature of the coupling is reflected in the behaviour shown in the upper panels.

6.2 Simulating the Strange Situation Procedure

We next describe experiments with an agent-based simulation closer to the full robotic model described above and illustrated in figure fig:architecture. In this simulation, point agents representing the child and caregiver are placed in a square arena, and a central base is fixed for the caregiver agent, similar to the chair often provided for the human caregiver in the SSP, allowing only small movements around this location. The child agent is able to choose between one of the four corners to explore or the central base to approach. The physical distance is measured as the Euclidean distance in the simulated arena. The emotional distance is simulated using binary pulses in the approach stage of each agent.

Results are shown in figure 9. Here the left column shows agent trajectories in the simulated arena, the central column the distance between child and care-giver agents

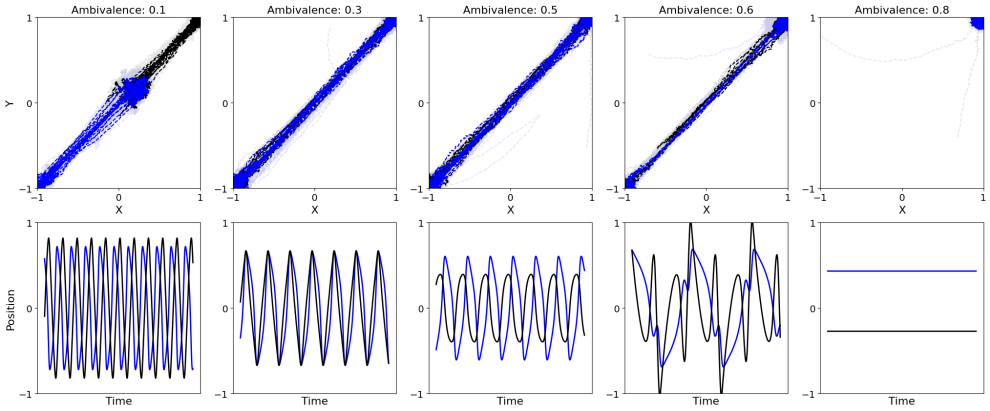


Fig. 7 Secure-to-ambivalent transition. Same as in figure 6. The parameter ϵ_{Am} increases from left to right.

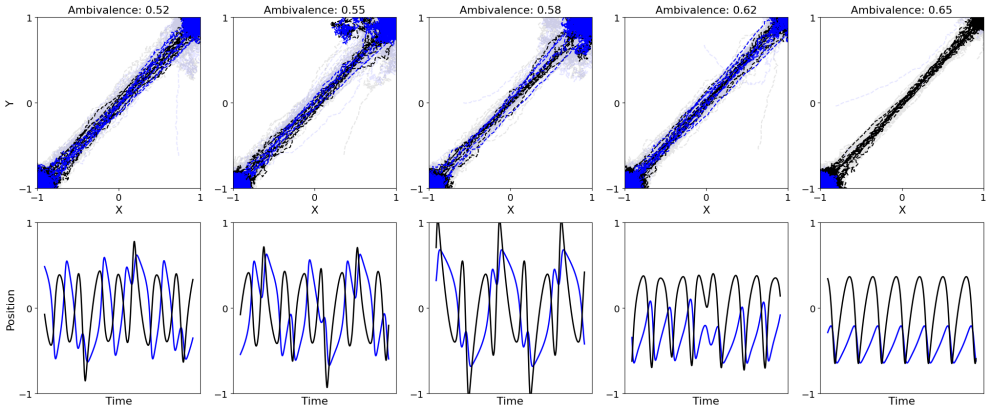


Fig. 8 Putative disorganised attachment. For a small parameter range, the model displays chaotic behaviour. Layout as in 6

and their vocalization behaviour, and the right column shows the agent's needs. The rows correspond to different $(\epsilon_{Av}, \epsilon_{Am})$ parameter regimes selected on the basis of the findings from the disembodied model (figures 3 and 4). Specifically, we show results (top to bottom in the figure) corresponding to the following attachment scenarios: putative secure $(0.0, 0.1)$, moderately avoidant $(0.4, 0.0)$, strongly avoidant $(0.9, 0.0)$, moderately ambivalent $(0.0, 0.4)$, and strongly ambivalent $(0.0, 0.9)$ regimes.

The top row in figure 9 illustrates a typical secure behaviour where vocalizations are aligned between the caregiver and the child, and the approach and explore behaviours are balanced. The next two rows illustrate moderately and strongly avoidant behaviour in which the child increasingly ignores the caregiver and spends an increasing proportion of its time exploring away from the base. Finally, the bottom two rows illustrate

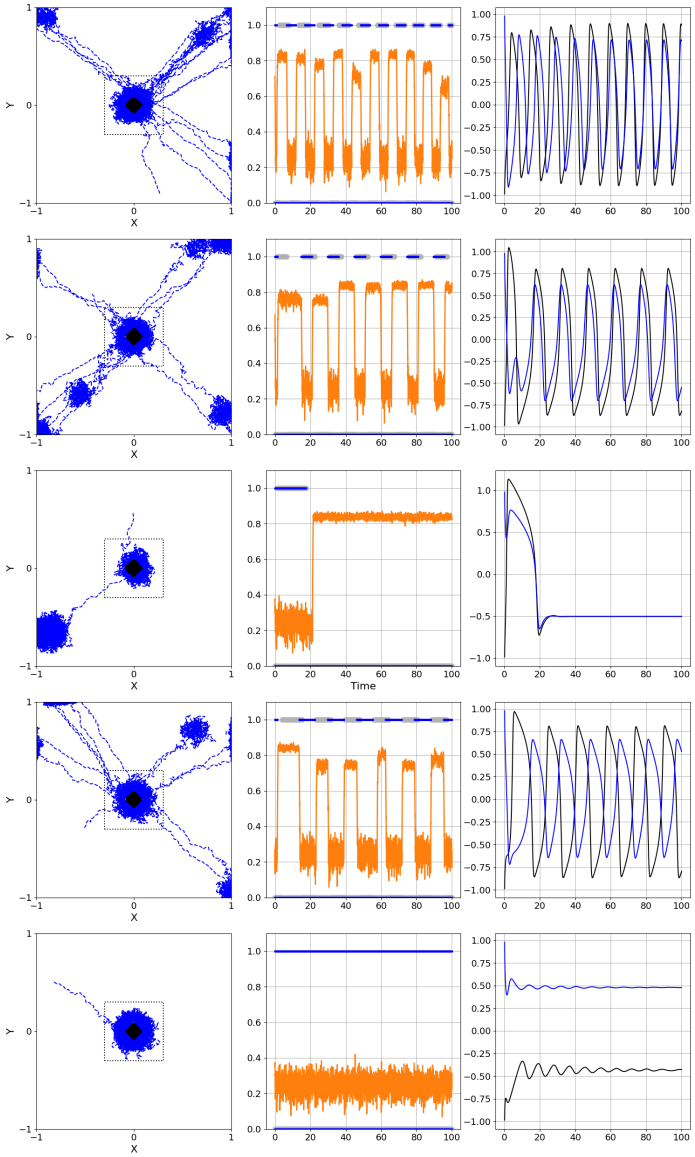


Fig. 9 Simulated strange situation task. **Left. Agent trajectories** within the simulated arena—blue: child, black: caregiver. **Centre. Inter-agent distance and vocalisation**. The distance is shown in orange and the vocalisations in blue (child) and gray (caregiver). **Right. Simulated needs** of parent and child over time. **Rows** show different combinations of parameters $(\epsilon_{Av}, \epsilon_{Am})$. From top to bottom: secure $(0.0, 0.1)$, moderate avoidance $(0.4, 0.0)$, strong avoidance $(0.9, 0.0)$, moderate ambivalence $(0.0, 0.4)$, strong ambivalence $(0.0, 0.9)$. The secure scenario shows the expected oscillation with child and caregiver receiving and giving care when needed. The strongly avoidant scenario shows behaviour where the child does not approach the parent (no oscillation). Finally, the moderately ambivalent scenario generates oscillations but they are not fully synchronised leading to slower satisfaction of the needs and a slowing of the oscillatory pattern.

moderately and strongly ambivalence behaviour. Here the child agent spends more time close to the caregiver with increasing strength of the ambivalence parameter. In the moderately ambivalent case, the requests for care and care-giving are not fully synchronised and therefore the needs are met more slowly and there is a slower oscillatory pattern. Slowing is also seen in the moderately avoidant scenario although without the desynchronisation. In the strongly avoidant/ambivalent scenarios the pattern of coupled oscillation breaks down and the system converges on the fixed attractor of continuous exploration/approach behaviour. In both cases this is consistent with findings from the disembodied model (Figure 3a) which predict slowing down as a phase transition is approached.

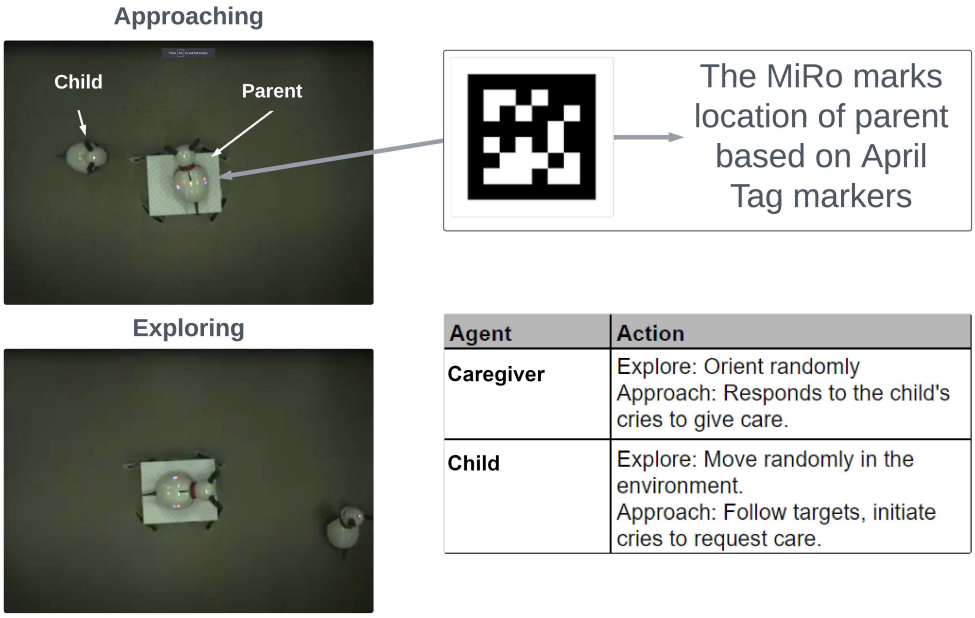
To investigate the differences in time spent close to or away from the caregiver, we calculate the proportion of total simulation steps during which the child agent is inside the dotted square shown in the leftmost panels. Consistent with the observations above, across the five scenarios this value was highly variable: secure 52%, moderately avoidant 41%, strongly avoidant 18%, moderately ambivalent 52%, and strongly ambivalent 99.9%. A Chi-squared test ($\tilde{\chi}^2 = 14424.6$, df= 4, N= 50,000, p¡0.001) confirmed significant differences in the amount of time spent close to the caregiver across the five regimes. Post-hoc Bonferroni-corrected z-tests show that all five regimes differ from each other at the $p < 0.05$ level, except for the secure and moderately ambivalent regimes where, in both cases, the child spends around half its time close to the caregiver.

6.3 Attachment-like behaviour in a robotic model

In a final series of experiments we used two MiRo-e robots—one representing the parent/childgiver and other the child—to investigate the full embodied model of the SSP. The robots were placed in an enclosed square arena with wide field camera recording their interaction and with a reflective marker on the head of the child robot for tracking (see figure 10). The caregiver robot was located in a box surrounded by April tags that the child robot could easily detect. The caregiver attends to the child as part of its exploration routine but giving/receiving of care was based solely on vocalizations (as explained earlier).

Based on the earlier simulation results we tested parameter regimes for putative secure, strong avoidance and moderate ambivalence. We selected a high level of ambivalence to test for the presence of the point attractor and a moderate level of ambivalence to be before the bifurcation point at $\epsilon_{Av} = 0.5$, but at a level where the oscillators are clearly desynchronised and the predicted slowing down is observed. For each scenario we placed the caregiver robot in the centre of the arena and the child robot in a random starting position away from the walls. We initialised the oscillator models with the required parameters and so that they would start in phase and ran the system for approximately five minutes.

Results for example runs for the different regimes are shown in Figure 11, robot trajectories within the arena are shown in the left-hand panels and the inter-robot distance and child robot vocalisations are shown in the right-hand panels. During these experimental runs it became clear that the child robot was occasionally getting stuck



Agent	Action
Caregiver	Explore: Orient randomly Approach: Responds to the child's cries to give care.
Child	Explore: Move randomly in the environment. Approach: Follow targets, initiate cries to request care.

Fig. 10 Experimental setup of the robot-robot SSP. The arena is an enclosed region in the robotics lab with a box in the middle, on top of which sits the caregiver. The child is tracked with a reflective marker. The box is surrounded with April tag markers that serve as targets for the child's approach behaviour. Once the child is close to an April tag it starts asking for care by vocalising. The approach routine of the caregiver consists of orienting towards the child and responding to care requests. The exploration routine of the child consists of choosing random points in the arena to explore; for the caregiver, it consists of random orientations within the box. The table in the figure shows the different actions available to the caregiver and child robots.

next to wall or in a corner and was unable to free itself immediately and was therefore not able to execute actions generated by the attachment model. Video analysis showed that these periods could be identified from the experimental record by outlier analysis——-passages of time when the robot was in the same arena position for a duration that was greater than two standard deviations above the mean duration. These passages of behaviour were excluded from the analysis but are labelled in the plots.

Figure 11 (top) shows an example run for the secure regime. Here the child explores the arena and uses the caregiver as a secure base. Each caregiving interaction is successfully completed. Note that interactions are quite short and happen when the child is close to the caregiver. This behaviour matches that found in the simulation run for the secure regime (figure 9, top) where the model generated a regular pattern of interaction bouts alternating with exploration behaviour.

Figure 11 (middle) shows an example of system's behaviour in the avoidant regime. The child spends most of its time away from the parent having short bouts of vocalizations before switching back to exploration. Even when it is close to the base for a long time, the lack of vocalizations indicates an explore state. The behaviour of the

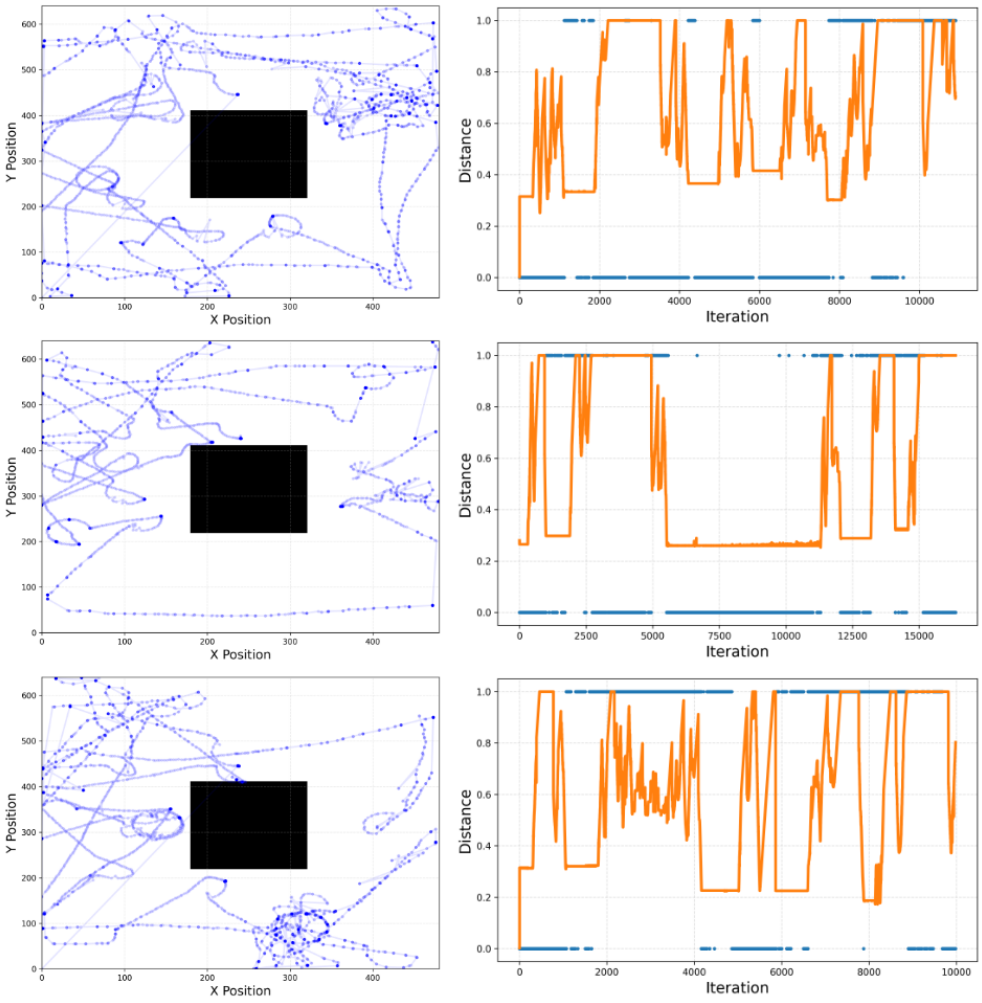


Fig. 11 Left. Trajectory of the child relative to the caregiver. **Right. Inter-robot distance and vocalisation**. Distance is from the centre of the arena normalized and plotted in orange, higher values indicate exploration bouts. Child vocalisations are shown in blue. **Top. A secure scenario** $\epsilon_{Am} = 0.1$ corresponding to a parameter regime in which both oscillators are synchronized. The child explores the space safely and returns to base regularly. In the secure regime, vocalizations occur close to the caregiver and are attended to immediately (as indicated by their length). **Middle. Avoidance scenario** $\epsilon_{Av} = 0.9$— The child spends increasing time far from the base. In the long region close to the base around iteration 7500, the lack of vocalizations indicates an exploratory state. **Bottom. Ambivalence scenario** $\epsilon_{Am} = 0.4$—the child spends most of the time far from the parent and the exploration bouts are punctuated by short care request attempts before switching back to exploration.

robot differs in some aspects from that in the simulation experiment (Figure 9, middle panel), for example, there appears to be some oscillatory. We return to this in the discussion.

Finally, figure 11 (bottom) shows an example of a moderately ambivalent robot dyad. Here the oscillators are out-of-phase and are close to the bifurcation. Calls for care are not reciprocated by the caregiver as indicated by the long vocalisation bouts without a successful care interaction. With respect to the physical distance, it spends about half of the time near the base. This is again similar to the simulation experiment (Figure 9, bottom).

To further investigate the behaviour of the child robot in the different parameter regimes, we calculated heat-maps of the most visited locations (figure 12) after removing the outliers noted above. We also divided the arena into regions close to and away from the caregiver, as in the simulated model, and calculated the proportion of total trial steps during which the child was inside the dotted square shown in the lefthand panels in figure 11. This analysis found the proportions of time spent close to the caregiver to be secure 54%, strongly avoidant 41%, and moderately ambivalent 43%. A Chi-squared test ($\tilde{\chi}^2 = 99.38$, df= 2, N= 29743, p¡0.001) showed significant differences in the amount of time spent close to the caregiver across the three regimes. Post-hoc Bonferroni-corrected z-tests show that the secure regime differed from the other two at the $p < 0.05$ level, but that strongly avoidant and moderately ambivalent did not differ from each other. We note that the difference between secure and strongly avoidant is in the expected direction with the avoidant robot spending more time away from the caregiver. However, that the ambivalent robot also spent more time away from the caregiver compared to the secure scenario, was not expected from the simulation model (figure 9) which showed no difference.

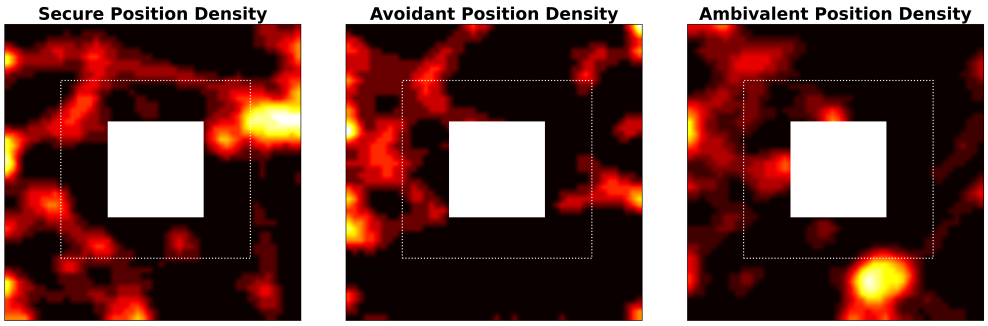


Fig. 12 Heat map visualization of agent position over time on a 50×50 grid. The colour magnitude reflects the frequency with which the agent is detected in each region of the arena. Outliers corresponding to parts of the experiment where the robot is stuck in a wall were removed. **Left**: Secure behaviour at coupling strength $\epsilon_{Am} = 0.1$, characterized by phase-locked dynamics. **Middle**: Avoidant behaviour at $\epsilon_{Av} = 0.9$. **Right**: Ambivalent behaviour at $\epsilon_{Av} = 0.4$. The results match qualitatively the simulation.

7 Discussion

Results from both the simulation and robotic models show patterns of oscillatory behavior, for the secure attachment scenario, with alternating bouts of exploration

and care-seeking, that resembles those found with human participants in the Strange Situation procedure (e.g. figure 1). We have also shown that parameters related to the attractive and repulsive coupling components within the model can be used to simulate ambivalent and avoidant attachment styles that are characterised by either excessive approach or avoidance of the care-giver. Additionally, just before the transition to a steady state, we observed non-periodic regimes (critical fluctuations), we also observed regions of chaotic behaviour for some parameter values. Future experimental work in developmental settings could potentially test whether similar patterns arise in parent-child interactions.

Interestingly, the patterns of distances and vocalizations, in our robot model, vary greatly for different parameter regimes, with episodes of low and high proportions of vocalizations as observed in human dyads when placed in the strange situation. Our dynamical model predicts clear regions of synchronicity, anti-synchronicity, chaos and steady states in the phase space. A limitation of the current model, that we are working to understand, is that some of the qualitative dynamics of the original dynamical model is not fully reproduced in our robotic implementation. For instance, high avoidance should lead to a point attractor (figure 3b, figure 10 middle), whereas we see some continued oscillation in the robot model (figure 12, left). This could be due, in part, to the nature of real-world robotics where motor behaviour can deviate from that expected given the control signals, and where sensor signals can be intermittent and noisy. This could lead to behaviour that escapes local attractors such as the fixed point attractor observed in the avoidant scenario due to external emotional and physical inputs. Future work will probe this difference and try to isolate the contributions of different aspects of the embodied system.

For the robotic implementation, we assumed that the agents have access to some information about the needs of the other agent to couple the oscillators. However, this is clearly not the case in any real social interaction. In such situations, the state or the coupling must be inferred from the information gained by observing the behaviour of the other agent. A reasonable approach would be to sequentially estimate these values from emotional and physical interactions, or at least to estimate the coupling functions given in equation 9. Alternatively, it is possible to leverage the oscillatory nature of the model to develop pulse-coupling approaches (e.g. [26]) that involve maintain an internal model of the other agent, keeping it synchronized with their behaviour, and then couple this model with the agent itself. We are currently working towards a demonstration of this idea.

A previous dynamical account provided by Stevens [27], sought to cast attachment within the framework of homeostasis and modelled the development of attachment as a long-term process of adaptation. In contrast, the focus here is on short-term adaptations within time frames comparable to Bowlby's [1] original discussions of patterns of approach and exploratory behaviour in caregiver-child dyads. The oscillator model developed here is particularly useful in capturing this kind of interaction. Nevertheless, we are also interested in exploring the longer-term changes in attachment dynamics and their links to underlying drives concerned with survival needs. In [15] we recently reviewed some dynamical and modelling research in this direction. The ultimate goal

would be to link models of long and short dynamics, using fast-slow dynamical models. For instance, long-term dynamics could be used to model changes in some of the fixed parameters in the current model such as those relating to attachment style. Ultimately, we should expect that these could change gradually with time in part due to the interaction history of the dyad.

The use of robotics creates a focus on operationalising theories of coupling between agents while also generating a behavioural richness that can be absent in disembodied models. There are interesting parallels here with earlier work by Canamero et al. [28] who developed a model of imprinting in a physical robot. The robot we have used is animal-like rather than human-like, partly as our broader aim is understand the mammalian brain architecture for emotional and social behaviour. From this perspective human attachment can be seen as the expression of an underlying layered motivational control system that is present in all mammals [15, 29, 30]. We are also interested to explore similarities between attachment and other mammalian social behaviours such as filial huddling [30, 31].

People can form emotional bonds with intelligent artefacts, such as social and companion robots, that are complex and can be personally meaningful [32]. Some forms of bonding could be beneficial in situations where the robot provides a valued service such as being physically- or socially- assistive or have a therapeutic role. At the same time, concerns have been raised that inappropriate bonding with a robot could cause harm (e.g. [33]). The nature of the behavioural interaction between human and robot is likely to be a determining factor in establishing any long-term bond. We have previously analysed human-robot relationships through the lens of social psychology and relationship science, proposing that there can be similarities between human-robot bonds and some forms of human-human, human-animal and human-object bond [32, 34]. Robillard and Hoey [35] have proposed that assistive technologies, such as robots, can usefully align their emotional expression with the emotional state of the user. We consider that the ability to estimate human attachment style and to generate appropriately-matched robot behaviours, as we have begun to explore here, could suggest methods for implementing better emotional alignment in social robot behavior. In the longer term this could lead to advances in the development of assistive robots.

8 Conclusion

We have presented a dynamical systems model of attachment—the primary bond between a human child and their caregiver—based on the novel proposal human attachment can be viewed as the behavioural coupling of two oscillators. In simulation and robot experiments, we found that this oscillator framework can generate rich patterns of behaviour that can be compared with observations of the Strange Situation procedure, an experimental paradigm that is widely studied in human relationship science. Moreover, following a dimensional view of attachment, we have shown that parameters within the model can be adjusted to capture some of the characteristic features of human avoidant and ambivalent attachment styles. By embodying the model in a robot form we were able to show that the coupled oscillator model can generate rich patterns of approach and exploration behaviour, regulated by the robot's internal

model of its own and its partner's needs, and was able communicate these needs, and maintain behavioural synchronisation via vocalisation. We hope that this simplified dynamical model of attachment could serve as useful basis for further understanding of human-human attachment behaviour. The use of dynamical models of self and other could also be developed in the future as a way to endow robots with improved capacities for social interaction.

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10 Declaration of Interest

TJP is a director and shareholder of Consequential Robotics Ltd which develops the MiRo robot, and Bettering Our Worlds (BOW) ltd which develops robot software. Other authors have no competing interests.

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